

Task 2: Smart Order Routing (SOR)

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BlockHouse Trial Task – Quantitative Research

1. Introduction and Motivation

In fragmented equity markets, the execution of even a single trade can span multiple exchanges, each with its own fee structure, queue state, and liquidity characteristics. The paper *Optimal Order Placement in Limit Order Markets* by Cont and Kukanov (2014) addresses a fundamental problem faced by institutional and high-frequency traders: given an order of size S to be executed within a short time horizon T , how should it be optimally allocated across limit and market orders, and across multiple venues?

Unlike earlier theoretical models which assume stylized market conditions or a single execution venue, the authors develop a realistic, data-driven framework to minimize the total expected execution cost. This cost includes explicit trading fees and rebates, market impact, and penalties for execution risk. The resulting optimization problem is convex and tractable, providing both analytical solutions (for single exchange) and stochastic algorithms (for multi-venue settings).

2. Formulation of the Optimization Problem

The order placement problem is cast as minimizing the expected cost:

$$V(X) = \mathbb{E}[v(X, \xi)],$$

where $X = (M, L_1, \dots, L_K)$ denotes the size of the market order M and limit orders L_k on K venues, and ξ represents random order outflows (cancellations and marketable orders) on each venue during $[0, T]$.

The total cost $v(X, \xi)$ comprises:

- Execution costs: market order fees, limit order rebates, and penalties for adverse selection.
- Execution risk: soft penalties for overfill or underfill based on the deviation from the target S .
- Market impact: modeled as a linear cost on submitted volume.

The model respects execution mechanics (FIFO for limit orders), queue sizes, and outflow realizations. Importantly, it incorporates fill randomness via ξ , allowing execution risk to be formally quantified.

3. Market vs Limit Order Allocation

For a single exchange ($K = 1$), the authors derive an explicit condition for the optimal split (M^*, L^*) . The result hinges on the trade-off between the immediacy of market orders (high cost, no risk) and the uncertainty but cheaper nature of limit orders (rebates, but execution risk). As execution urgency λ_u increases, the optimal policy shifts toward greater use of market orders. Conversely, generous limit order rebates or thin queues promote limit order usage.

The analytic solution shows that $M^* + L^* = S$, but their composition depends critically on queue size Q , expected outflows, and penalty parameters λ_u and λ_o . The model also confirms that the tail of the outflow distribution ξ is more influential than its mean—highlighting the importance of extreme events in execution planning.

4. Multi-Exchange Routing and Overbooking Behavior

For $K > 1$ exchanges, the optimization becomes higher-dimensional and must be solved numerically. The authors propose a stochastic approximation algorithm based on Robbins-Monro updates. This algorithm adapts allocations using historical fill outcomes, and exhibits convergence guarantees under mild assumptions.

An important empirical observation is the phenomenon of overbooking—placing total orders exceeding S —to mitigate execution risk via partial fills across venues. This rationalizes observed ‘phantom liquidity’ in real markets, where limit orders are submitted and quickly canceled. The model predicts that as K increases, traders rely less on market orders and more on dispersed limit orders, consistent with empirical depth observations.

5. Role of Market Microstructure Inputs

Microstructure variables such as queue size Q_k , fee/rebate structures f_k and r_k , and historical outflow distributions ξ_k play pivotal roles in optimal allocation. The queue position determines fill probability, while the fee structure directly shifts cost tradeoffs. The authors also show that under low queue correlation and moderate cost asymmetries, traders prefer to spread limit orders, exploiting statistical independence of fills.

Their empirical backtest on MSFT tick data further confirms that optimal solutions substantially reduce execution cost compared to naive strategies (e.g., equal split or pure market orders), especially for moderate order sizes.

6. Conclusion

Cont and Kukanov’s framework presents a realistic and theoretically grounded approach to smart order routing. By decoupling scheduling from placement and explicitly modeling execution uncertainty, it offers a robust foundation for high-frequency trading systems. Understanding queue dynamics, market fragmentation, and the stochastic nature of fills is essential for practitioners optimizing execution quality in today’s fragmented markets.